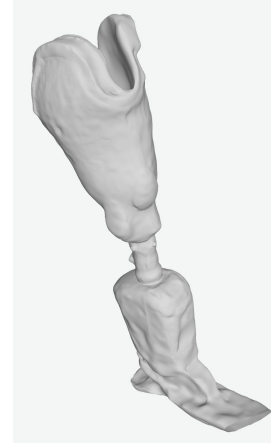


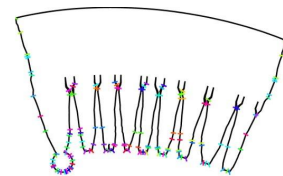
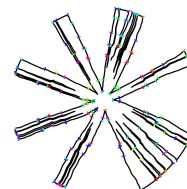
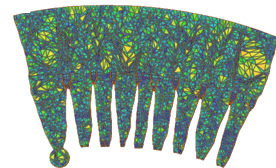
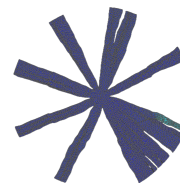
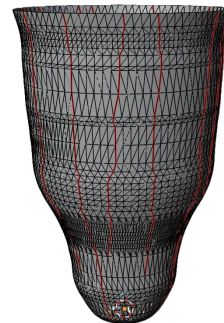
Photogrammetry Scanning

- Used Kiri Engine, a smartphone-based photogrammetry/LiDAR scanning app
- 150 circumferential images were taken of the participant's extended limb in a seated position
- Post-processing in Blender



Seam-Placement and Unraveling

- “Seams to Sewing Pattern” add-on in Blender for flat-pattern generation
- Calculated discrete Gaussian curvature per-vertex and assigned each point a K value where top nth percentile of $|K|$ are flagged as cut candidates
 - $K = 2\pi - \sum \theta_i$
- Iterates through points weighing distance and value to avoid clustered cuts
- Points are connected to create seams with single-sheet constraint



Residual Limb Model

- 3D printed a life-size model in PETG and 1:3.5 scaled model in PETG and TPU
 - Coated the life-size model in epoxy and silicone
- Simulate increased temperature using circulated heated water
- Prototype liner design ideas before full-scale fabrication

